

S V V Sai Sri Vallabh Pataneni

Master's Thesis: Process Engineering, Test-Rig & Thermal Characterisation

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SUMMARY

An Automotive Engineering Master's student at RWTH Aachen who enjoys building test rigs and characterising thermal behaviour. At Ecogenium I engineered and instrumented a standalone fuel-cell test bench from scratch, and at DLR I built a thermal-management model to specify coolant flow, temperatures and a heat exchanger for an aircraft fuel-cell system. I design in SolidWorks and CATIA V5. I am seeking the Master's thesis at Reverion to build and commission a pressure test chamber and characterise the thermal conductivity of insulation materials for your reversible high-efficiency power plants.

KEY SKILLS

CAD & manufacturing: SolidWorks · CATIA V5 · Siemens NX · metal 3D printing (DMLS) · DFM/DFAM · production engineering

Test & measurement: test-rig build & commissioning · instrumentation · polarisation curves · thermal-window mapping

Simulation: MATLAB/Simulink (MIL) · ANSYS (Fluent, Structural) · Star-CCM+

Programming & data: Python (NumPy, Pandas) · C/C++ · Power BI

PROFESSIONAL EXPERIENCE

Mechanical / Suspension Team Lead & Pit Crew, Ecogenium e.V (RWTH student team) · 06/2024 – Present

Aachen, Germany

- **Fuel-Cell Test Bench:** Engineered, fabricated and electronically integrated a standalone fuel-cell test bench, running purging, humidity cycling, short-circuiting and polarisation-curve experiments and mapping the stack's optimal thermal operating window.
- **Manufacturing & Test Equipment:** Built and commissioned test rigs and fixtures, designed parts in CATIA V5 / Siemens NX, and prototyped via metal 3D printing (DMLS) with hands-on DFM/DFAM.
- **Energy-Management Strategy:** Architected and deployed energy-management and fuel-cell temperature-control strategies in Python/Simulink (MIL-validated), boosting on-track fuel efficiency by ~80% and securing a runner-up finish at Shell Eco-Marathon 2025.

Werkstudent, Fuel-Cell Systems (Aircraft), DLR Innovation Center for Small Aircraft Technologies (INK) · 01/2026 – Present

Würselen (Aachen), Germany

- **Thermal-Management Digital Twin:** Built a MATLAB/Simulink digital twin of the complete thermal-management system to derive design specifications (volumetric flow, coolant temperature range, coolant pressure), driving procurement of the coolant pump and heat exchanger.
- **Component Selection:** Selected every component within the thermal-management system against the derived specifications.
- **System Rig Design:** Designed a compact, aircraft-engine-sized fuel-cell system rig in CATIA V5, optimised for easy transferability.

Operating Strategy / System Simulation Intern, Robert Bosch GmbH · 09/2025 – 12/2025

Schwieberdingen, Germany

- **Analysis & Documentation:** Built post-processing and KPI analysis (power distribution, SOC window, efficiency, degradation) and co-authored a peer-reviewed paper on the rule-based FCEV control strategy.

EDUCATION

M.Sc. Automotive Engineering, RWTH Aachen University · 09/2023 – 06/2026

Aachen, Germany · GPA 1.7 (1.2 in core automotive courses) · Fuel-Cell System Technology, Alternative/
Mobile Propulsion, Simulation Sciences, FEM, CFD

B.Tech Mechanical Engineering, JNTU Hyderabad · 06/2019 – 04/2023

Hyderabad, India · GPA 1.3 (German scale)

LANGUAGES

English (Fluent, C1) · German (Intermediate, B1) · Telugu / Hindi (Native)